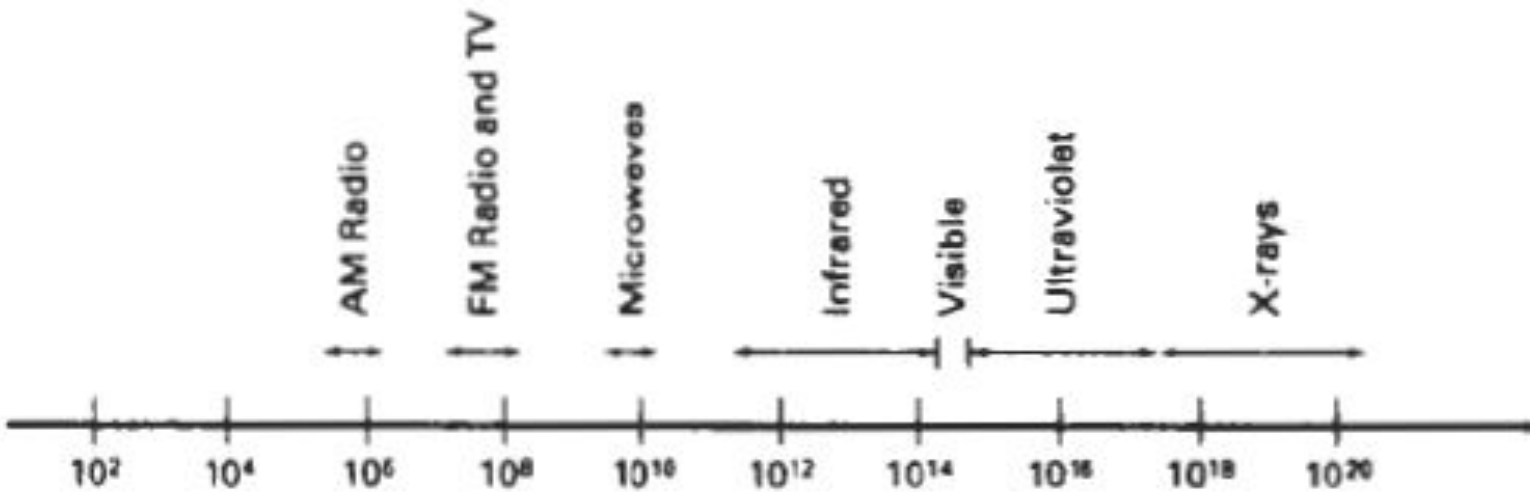
Module 4

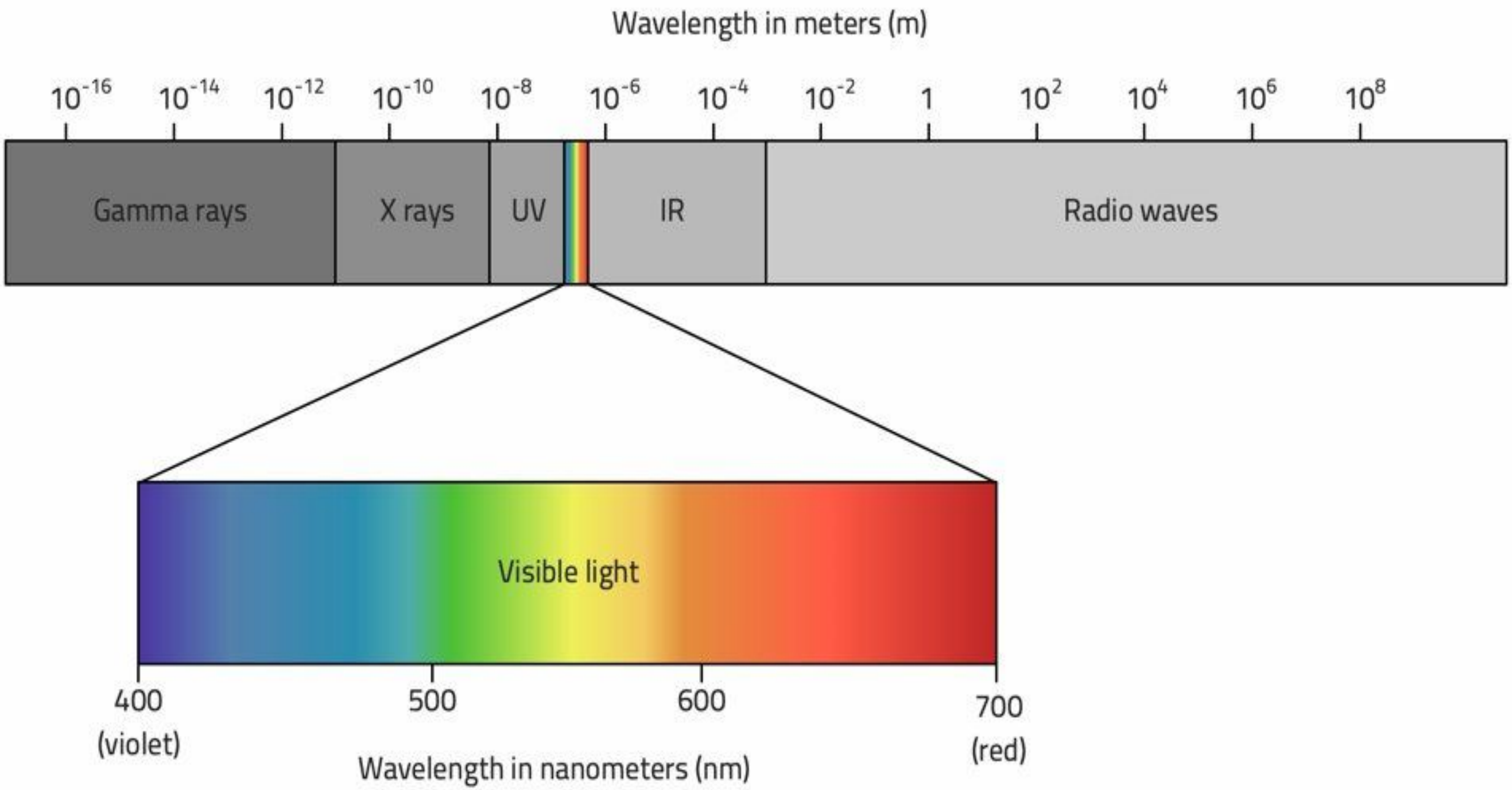
Color models and Color applications

- A color model is a method for explaining the properties or behavior of color within some particular context.
- No single color model can explain all aspects of color, so we make use of different models to help **describe** the different perceived characteristics of color.

PROPERTIES OF LIGHT

- Color is electromagnetic radiation within a narrow frequency band
- A few of the other frequency bands within this spectrum are called radio waves, microwaves, infrared waves, and X-rays.



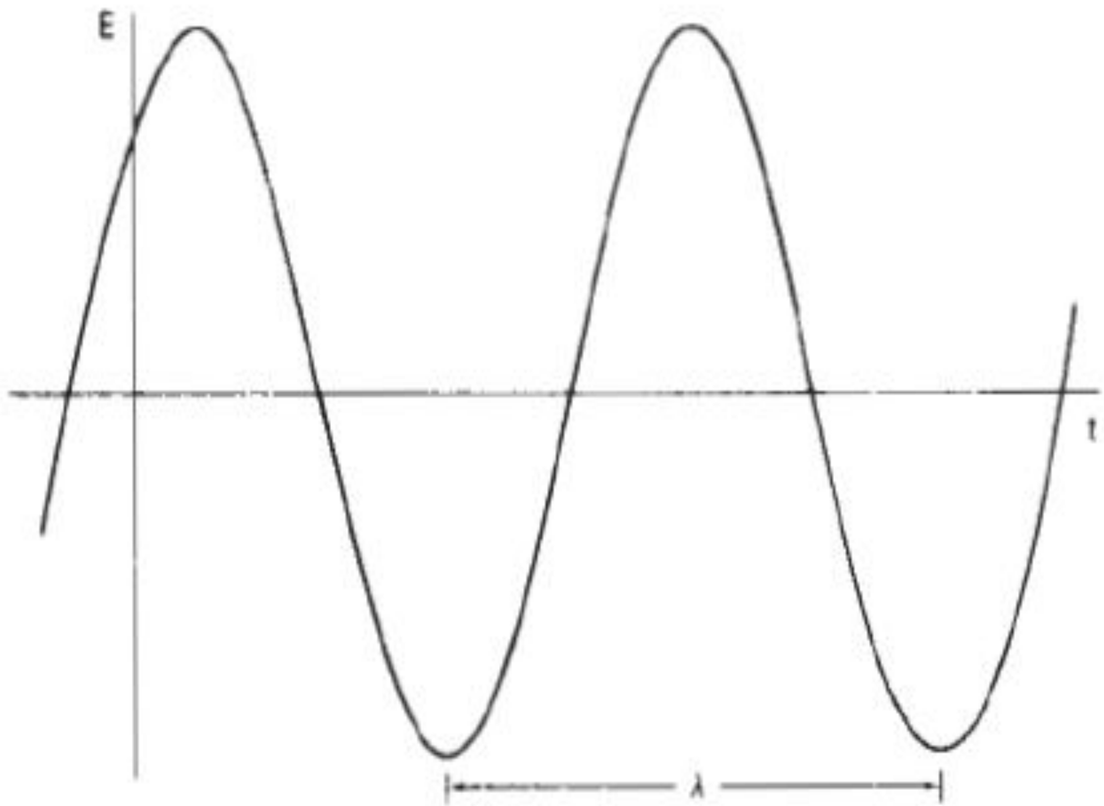


Wavelength in meters (m)
10^{-16} 10^{-14} 10^{-12} 10^{-10} 10^{-8} 10^{-6} 10^{-4} 10^{-2} 1 10^{2} 10^{4} 10^{6} 10^{8}
Gamma rays
X rays
UV
IR
Radio waves
Visible light
400
(violet)
500
600
700
(red)
Wavelength in nanometers (nm)

- Each frequency value within the visible band corresponds to a distinct color.
- At the low-frequency end is a red color (4.3×10^{14} Hz), and the highest frequency we can see is a violet color (7.5×10^{14} Hz).
- Spectral colors range from the reds through orange and yellow at the low-frequency end to greens, blues, and violet at the high end

- Light is an electromagnetic wave emitted by a source.
- Light is characterized by
  – Frequency(f)
  – Wavelength(λ)
- The wavelength and frequency of the monochromatic wave are inversely proportional to each other, with the proportionality constant as the speed of light **c:**
  – **Speed c=f x λ**
  – **C=3x** $10^8 m/s$ **in vacuum**
  – Frequency is constant for all materials

- Light wavelengths are very small, so length units for designating spectral colors are usually either angstroms **(1** $\mathring{A}$ = 10^{-8}cm) or nanometers (1 nm = 10^{-7}cm)
- An equivalent term for nanometer is mill-micron.
- Light at the red end of the spectrum as a wavelength of approximately 700 nanometers (nm), and the wavelength of the violet light at the other end of the spectrum is about 400 nm.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λ

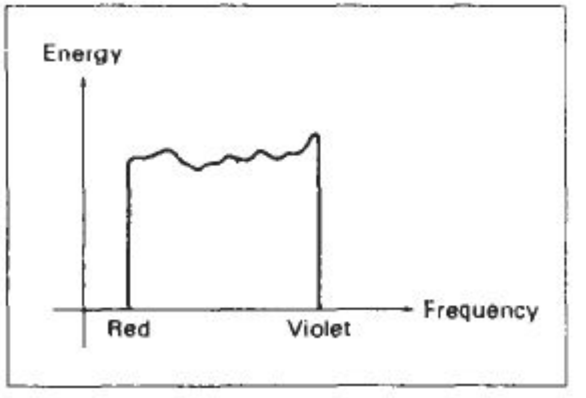
Components of light

- A light source such as the sun or a light bulb emits all frequencies within the visible range to produce white light.
- When white light is incident upon an object, some frequencies are reflected and some are absorbed by the object.
- The combination of frequencies present in the reflected light determines what we perceive as the color of the object.
- If low frequencies are predominant in the reflected light, the object is described as red.

- The perceived light has a dominant frequency (or dominant wavelength) at the red end of the spectrum.
- The dominant frequency is also called the **hue**, or simply the color, of the light (wavelength of light reaching the eyes after getting reflected from the object).
- When we view a source of light, our eves respond to the color (or dominant frequency) and two other basic sensations.
- One of these we call the **brightness/ luminance**, which is the perceived intensity of the light. Intensity is the radiant
- energy emitted per unit time, per unit solid angle, and per unit projected area of the source.

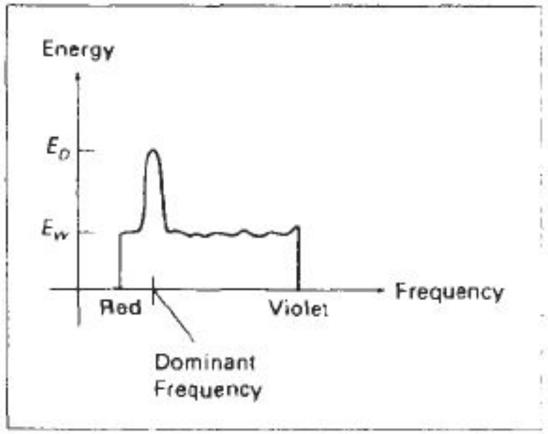
- Radiant energy is related to the **luminance** of the source.
- It is a measure of the visual sensation of brightness of the light emitted from a pixel.
- It does not depend on the surrounding.
- **Purity/ Saturation** describes how washed out or how "pure" the color of the light appears. Pastels and pale colors are described as less pure and dark has high purity.
- Saturation represent how much white is present in a color
- These three characteristics, dominant frequency, brightness, and purity, are commonly used to describe the **different properties** we perceive in a source of light.

- The term **chromaticity** is used to refer collectively to the two properties describing color characteristics: purity and dominant frequency/hue.
- Energy emitted by a white-light source has a distribution over the visible frequencies.



- Each frequency component within the range from red to violet contributes more or less equally to the total energy, and the color of the source is described as white.

- When a dominant frequency is present, the energy distribution for the source takes a form
- **We** would now describe the light as having the color corresponding to the dominant frequency.



- The energy density of the dominant light component is labeled as ***Ed*** in this ***figure,*** and the contributions from the other frequencies produce white light of energy density ***Ew.***
- We can calculate the brightness of the source as the area under the curve, which gives the total energy density emitted.

- Purity depends on the difference between E_D and E_W.
- The larger the energy E_D of the dominant frequency compared to the white-light component E_W, the more pure the light.
- We have a purity of 100 percent when $E_W = 0$ and a purity of 0 percent when $E_W = E_D$.

- Two different-color light sources with suitably chosen intensities can be used to produce a range of other colors.
- If the two color sources combine to produce white light, they are referred to as **complementar**y colors.
- Examples of complementary color pairs are red and cyan, green and magenta, and blue and yellow

- Typically, color models that are used to describe combinations of light in terms of dominant frequency (hue) use three colors to obtain a reasonably wide range of colors, called the **color gamut** for that model.
- The two or three colors used to produce other colors in such a color model are referred to as **primary colors**.

Standard Primaries and the chromaticity diagram

- International standard for primary colours proposed by Commission lnternational deI'Eclairage(CIE in 1931)
- The three standard primaries are imaginary colors.
- They are defined mathematically with positive color-matching functions that specify the amount of each primary needed to describe any spectral color.
- This provides an international standard definition for all colors, and the CIE primaries eliminate negative-value color matching and other problems associated with selecting a set of real primaries

XYZ Color Model

- The set of CIE primaries is generally referred to as the XYZ, or **(X, Y, Z),** color model, where **X, Y,** and Z represent vectors in a three-dimensional, additive color space.
- Any color C_λ, is then expressed as

$$C_\lambda = AX + BY + CZ$$

- where A, **B,** and C designate the amounts of the standard primaries needed to match C_λ

- Normalized amounts are thus calculated as,

  x = A/(A+B+C)

  y = B/(A+B+C)

  z = C/(A+B+C)

- $xX + yY + zZ$, where $x + y + z = 1$
- Thus, any color can be represented with just the ***x*** and **y** amounts.
- Since we have normalized against luminance, parameters **x** and y are called the ***chromaticity values*** because they depend only on hue and purity.

- To compute the actual amount from normalized amount
  - At lease one actual amount of one of the primaries is given
  - Two normalized amounts and actual amounts
- For example, x,y and B are known

$$\frac{x}{y} = \frac{A}{B}, \qquad A = \frac{x}{y} \bullet B$$

z=1-(x+y)

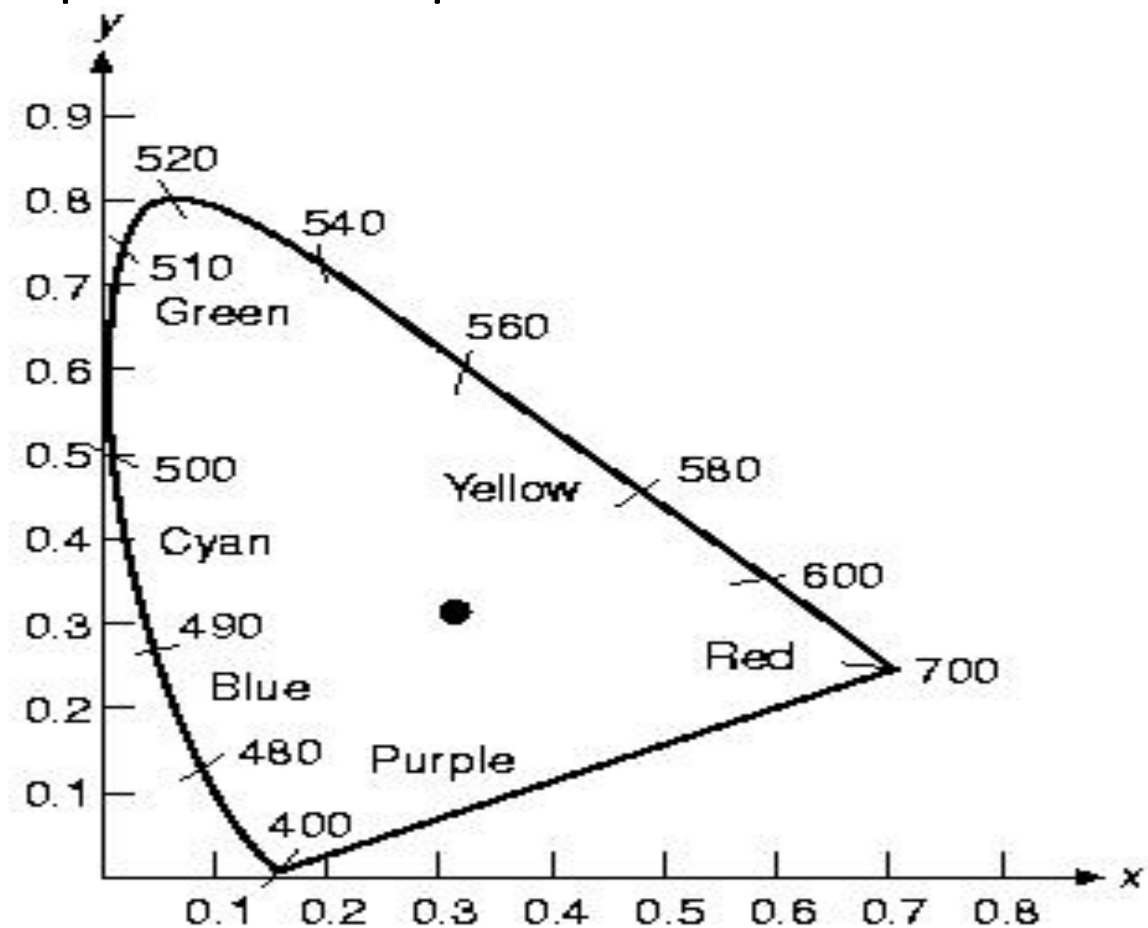
$$\frac{y}{z} = \frac{B}{C}, \qquad C = \frac{z}{y} \bullet B$$

CIE Chromaticity Diagram

- When we plot the normalized amounts **x** and y for colors in the visible spectrum, we obtain the tongue-shaped curve**.** This curve is called the **CIE chromaticity** diagram.
- Points along the curve are the "pure" colors in the electromagnetic spectrum labeled according to wavelength in nanometers from the red end to the violet end of the spectrum.
- The line joining the red and violet spectral points, called **the purple *line,*** is not part of the spectrum.
- Interior points represent all possible visible color combinations.
- Point *C* in the diagram corresponds to the white-light position.
- Actually, this point is plotted for a white-light source known as illuminant **C** which is k e d as a standard approximation for "average" daylight.

Luminance values are not available in the chromaticity diagram because of normalization. Colors with different luminance but the same chromaticity map to the same point.

Saturation



Hue

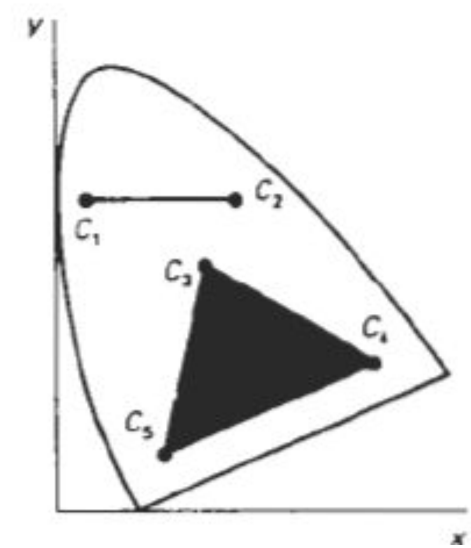
Uses of chromaticity diagram

- Comparing color gamuts for different sets of primaries.
- Identifying complementary colors.
- Determining dominant wavelength and purity of a given color.

Comparing color gamuts for different sets of primaries

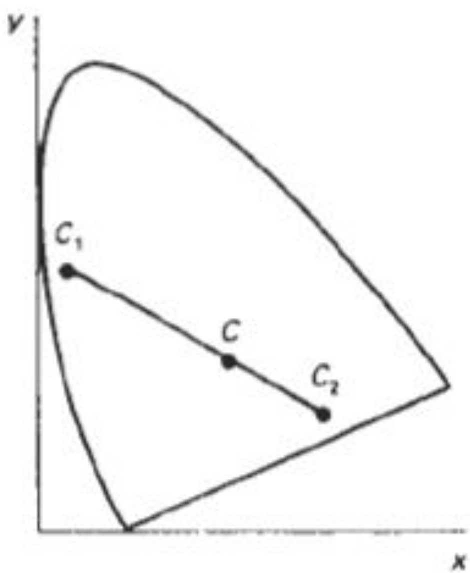
- Color gamuts are represented on the chromaticity diagram as straight line segments or as polygons.
- All colors along the line joining points C1, and **C2** in can be obtained by mixing appropriate amounts of the colors C1, and C2.
- If a greater proportion of C1, is used, the resultant color is closer to C1, than to C2

- The color gamut for three points, such as C3, C4,, and C5, in a triangle with vertices at the three color positions.
- Three primaries can only generate colors inside or on the bounding edges of the triangle.



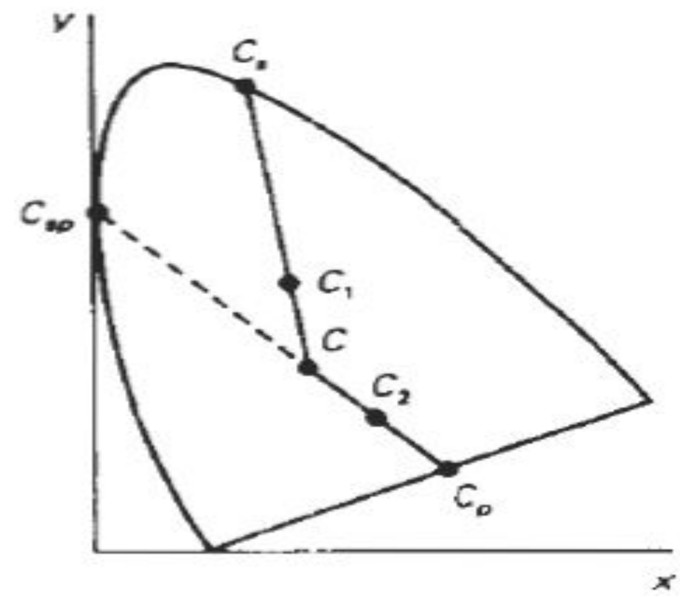
Identifying complementary colors

- The color gamut for two points is a straight line, complementary colors must be represented on the chromaticity diagram as two points situated on opposite sides of C and connected with a straight line.
- When we mix proper amounts of the two colors C1, and **C2**, we can obtain white light.



To determine the dominant wavelength of a color

- For color point C_1, we can draw a straight line from C through C_1, to intersect the spectral curve at point C_S,
- Color C_1, can then be represented as a combination of white light C and the spectral color C_S,



- Drawing a line from C through point C2 takes us to point Cp on the purple line, which is not in the visible spectrum.
- Point C2 is referred to as a ***nonspectral*** color, and its dominant wavelength is taken as the compliment of ***C_P,*** that lies on the spectral curve (point C_{SP},).

- For any color point, such as $\boldsymbol{C_1}$**,** we determine the purity as the relative distance of C_1, from C along the straight line joining C to C_S.
- If d_{C1}, denotes the distance from ***C*** to C_1, and d_{CS}, is the distance from C to C_S, we can calculate purity as the ratio $d_{C1}/\boldsymbol{d_{CS}}$.

COLOR MODELS

- There are two types of color models
  - Additive(RGB)
  - Subtractive (CMY)
- Additive color models use light to display and subtractive models use printing inks.
- Colours perceived in additive models are the result of transmitted light
- Colours perceived in subtractive models are the result of reflected light

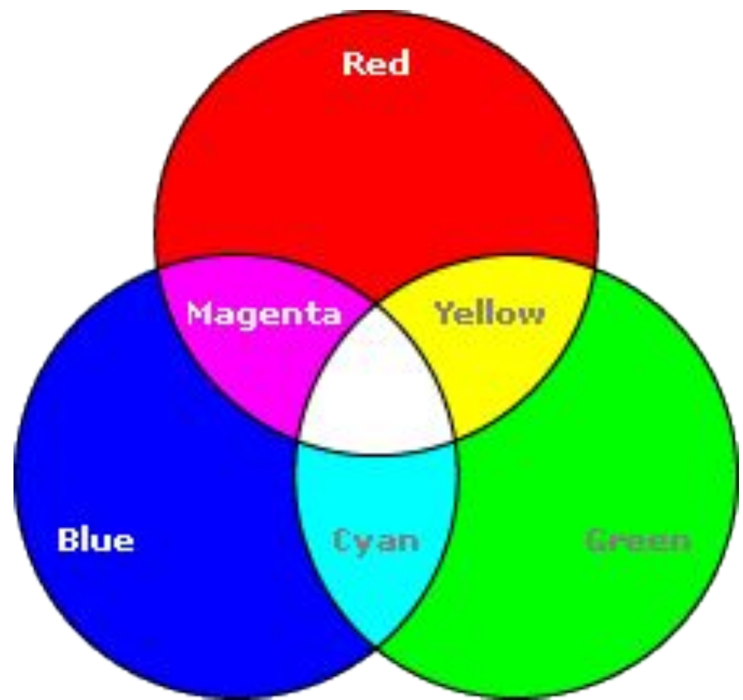
RGB COLOR MODEL

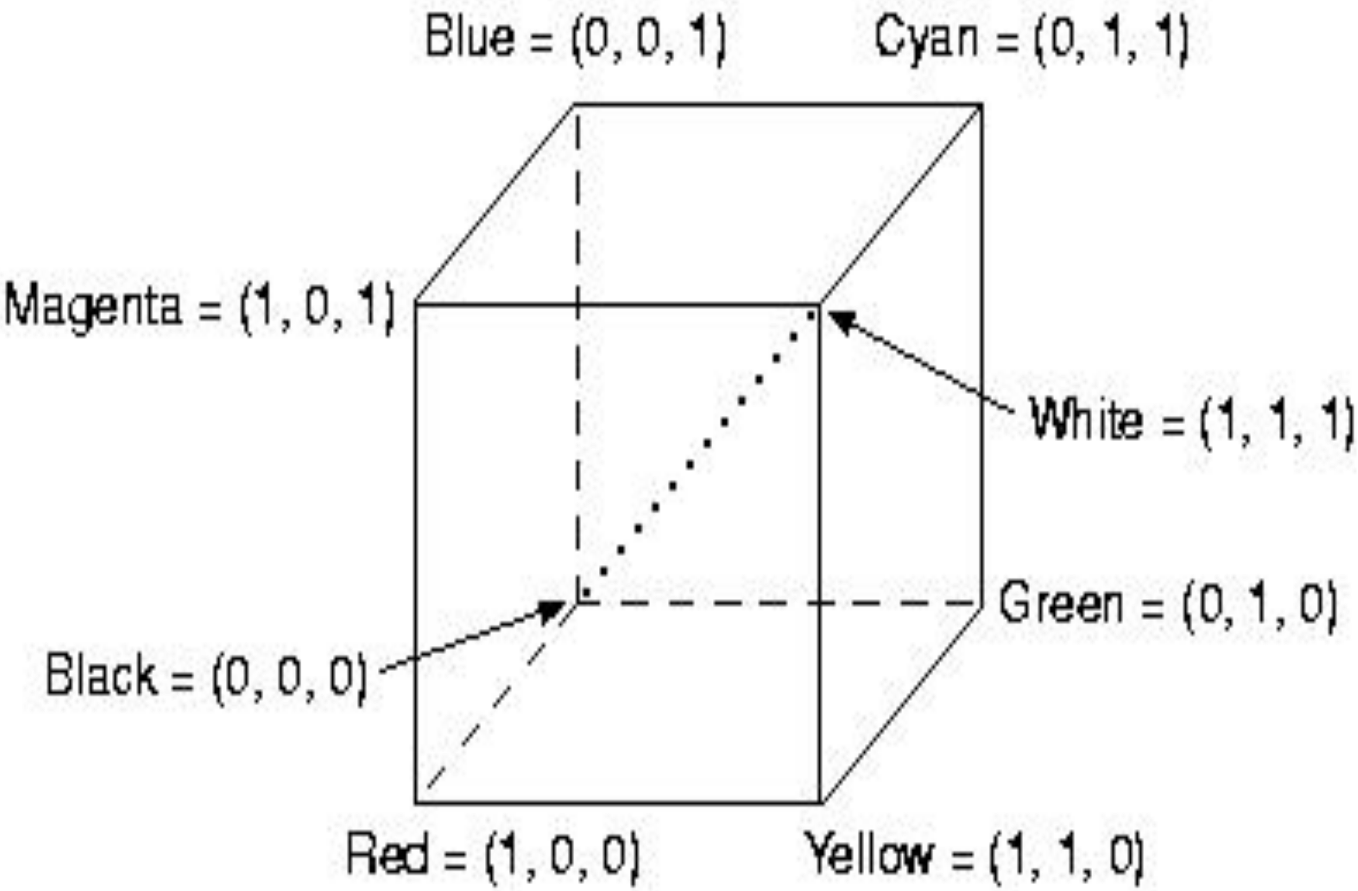
- RGB is an adaptive color model for computer display uses light to display color.
- Colors result from transmitted light.

  R+B+G=W

- We can represent this model with the unit cube defined on R, G, and B axes
- The origin represents black, and the vertex with coordinates (1, 1,1) is white.
- Vertices of the **cube** on the **axes** represent the primary colors, and the remaining vertices represent the complementary color for each of the primary colors

- $R = (1, 0, 0)$
- $G = (0, 1, 0)$
- $B = (0, 0, 1)$
- $R + G = (1, 0, 0) + (0, 1, 0) = (1, 1, 0) = Y$
- $R + B = (1, 0, 0) + (0, 0, 1) = (1, 0, 1) = M$
- $B + G = (0, 0, 1) + (0, 1, 0) = (0, 1, 1) = C$
- $R + G + B = (1, 1, 1) = W$
- $1 - W = (0, 0, 0) = \mathrm{BLK}$

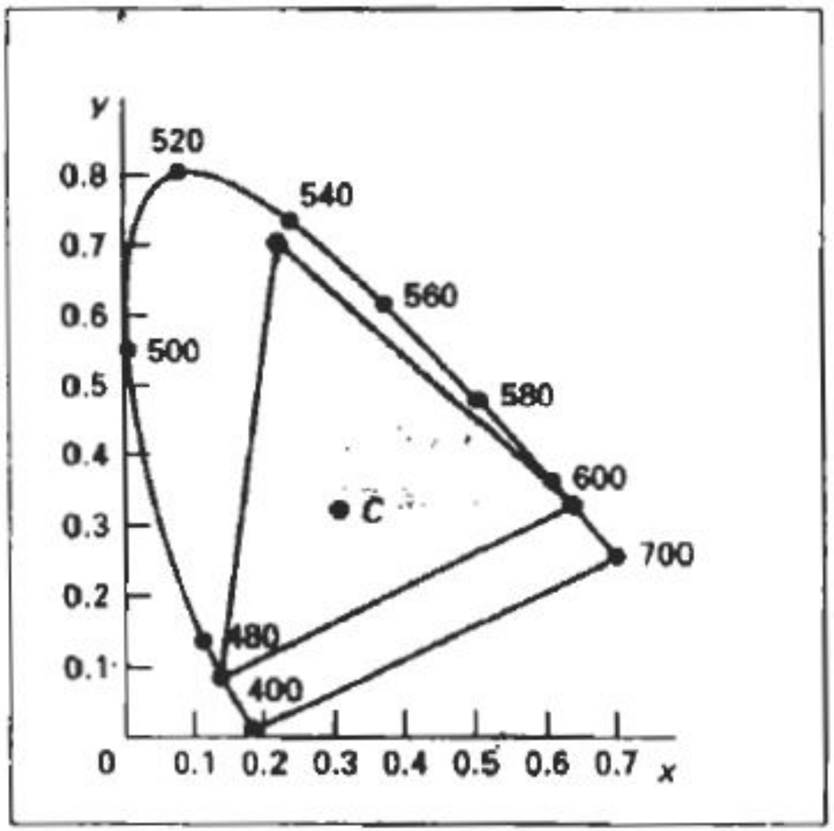



Blue = (0, 0, 1)
Cyan = (0, 1, 1)
Magenta = (1, 0, 1)
White = (1, 1, 1)
Green = (0, 1, 0)
Black = (0, 0, 0)
Red = (1, 0, 0)
Yellow = (1, 1, 0)

- Intensities of the primary colors are added to produce other colors.
- Each color point within the bounds of the **cube** can be represented as the triple ***(R,*** G, B), where values for R, G, and ***B*** are assigned in the range from 0 to 1.
- Thus, a color **Cλ,** is expressed in **RGB** components as

$$C_\lambda = R\mathbf{R} + G\mathbf{G} + B\mathbf{B}$$

- Shades of gray are represented along the main diagonal of the cube from the origin (black) to the white vertex.



RGB color gamut

YIQ COLOR MODEL

- The National Television System Committee (NTSC) color model for forming the composite video signal is the YIQ model.
- This is because the signals cannot be sent in RGB. So convert in the form of YIQ signal this will received by television monitor. Then convert into RGB format.
- This model was designed to separate chrominance(I & Q) and luminance(Y).
- The Y channel contains luminance(brightness) information while the I & Q channels carried the color information.

- In the YIQ color model, parameter Y is the same as in the XYZ model. Luminance (brightness) information is contained in the **Y** parameter, while chromaticity information (hue and purity) is incorporated into the ***I*** and Q parameters.
- A combination of red, green, and blue intensities are chosen for the **Y** parameter to yield the standard luminosity curve.
- Since **Y** contains the luminance information, black-and-white television monitors use only the *Y* signal.
- The largest bandwidth in the NTSC video signal (about 4 **MHz)** is assigned to the Y information.
- Parameter **I** contains orange-cyan hue information that provides the flesh-tone shading, and occupies a bandwidth of approximately 1.5 MHz.
- Parameter **Q** carries green-magenta hue information in a bandwidth of about 0.6 **MHz.**

- An RGB signal can be converted to a television signal using an NTSC encoder, which converts RGB values to YIQ values, then modulates and superimposes the I and Q information on the Y signal.
- The conversion from RGB values to YIQ values is accomplished with the transformation

$$\begin{bmatrix} Y \\ I \\ Q \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.144 \\ 0.596 & -0.275 & -0.321 \\ 0.212 & -0.528 & 0.311 \end{bmatrix} \cdot \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

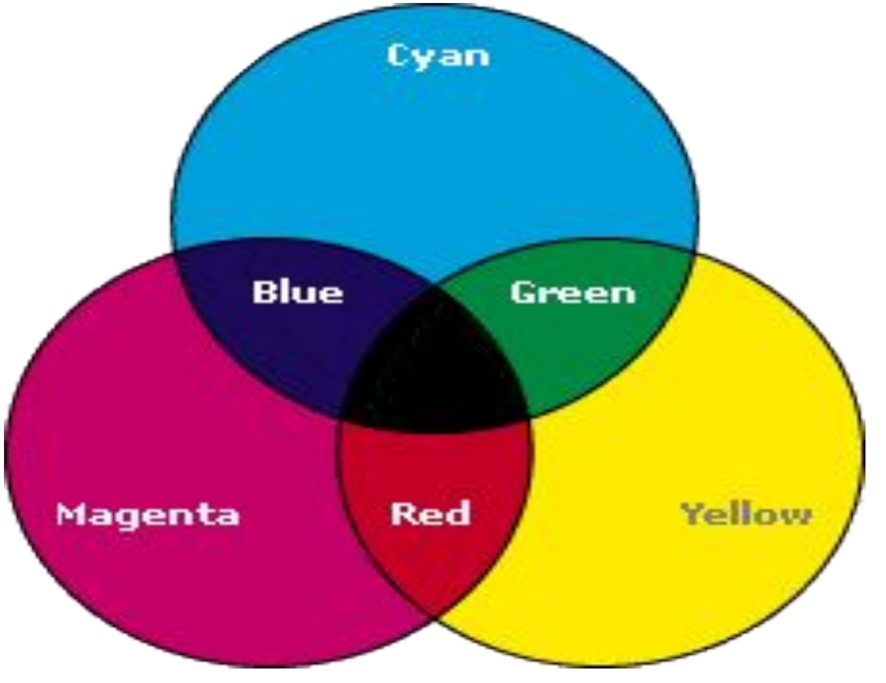
- An **NTSC** video signal can be converted to an RGB signal sing an NTSC decoder, which separates the video signal into the YIQ components then converts to RGB values.
- We convert from YIQ space to RGB space with the inverse matrix transformation

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 1.000 & 0.956 & 0.620 \\ 1.000 & -0.272 & -0.647 \\ 1.000 & -1.108 & 1.705 \end{bmatrix} \cdot \begin{bmatrix} Y \\ I \\ Q \end{bmatrix}$$

- Uses human's visual system properties to maximize information transmitted in a fixed bandwidth
  - *Y*: 4MHz
  - *I*: 1.5MHz
  - *Q*: 0.6MHz

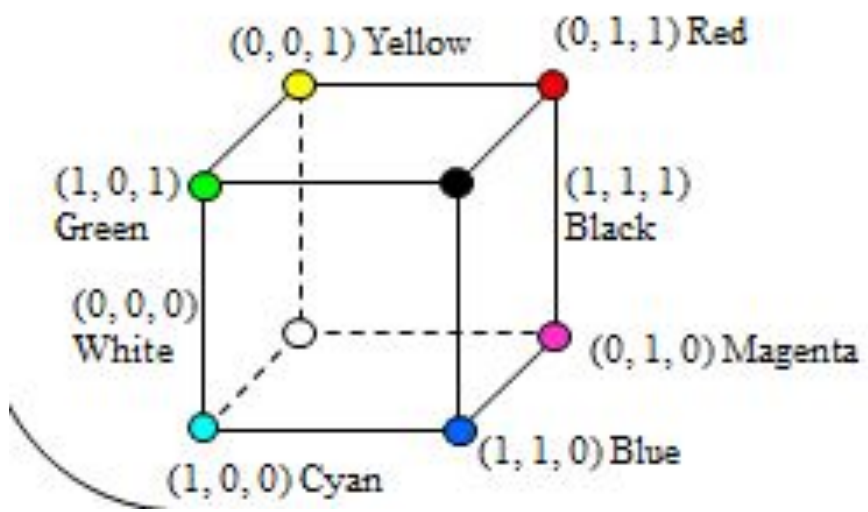
CMY COLOR MODEL

CMY: Complements of RGB



- **A** color model defined with the primary colors cyan, magenta, and yellow (CMY) is useful for describing color output to hard-copy devices.
- Cyan can be formed by adding green and blue light. Therefore, when white light is reflected from cyan-colored ink, the reflected light must have no red component.
    - C=G+B=W-R

- Magenta ink subtracts the green component from incident light, **Magenta**
    - **M=R+B = W-G**
- Yellow subtracts the blue component.
    - Y=R+G = W-B



- In the CMY model, point (1, 1, 1) represents black, because all components of the incident light are subtracted.
- The origin represents white light. Equal amounts of each of the primary colors produce grays, along the main diagonal of the **cube.**
- A combination of cyan and magenta ink produces blue light, because the red and green components of the incident light are absorbed.
- Other color combinations are obtained by a similar subtractive process

- The printing process often used with the **CMY** model generates a color point with a collection of four ink dots, somewhat as an RGB monitor uses a collection of three phosphor dots.
- One dot is used for each of the primary colors (cyan, magenta, and yellow), and one dot is black.
- A black dot is included **because** the combination of cyan, magenta, and yellow inks typically produce dark gray instead of black.

- Conversion from RGB to CMY

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = 1 - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

- where the white is represented in the RGB system is the unit column vector
- Conversion from CMY to RGB

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = 1 - \begin{bmatrix} C \\ M \\ Y \end{bmatrix}$$

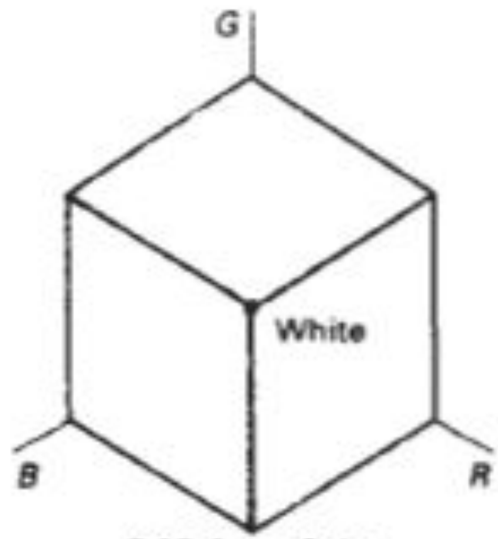
- where black is represented In the CMY system as the unit column vector.

HSV COLOR MODEL

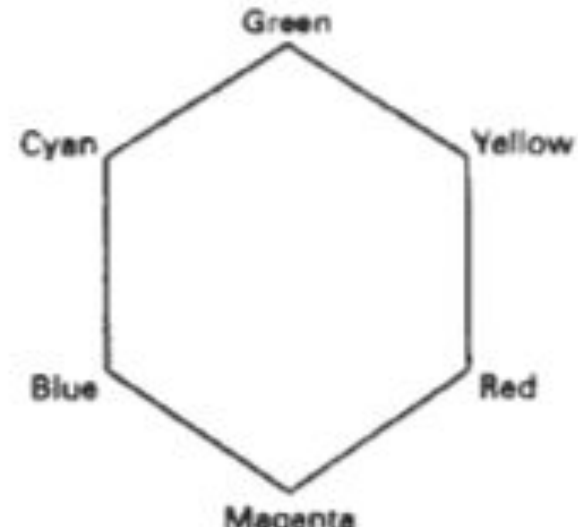
- lnstead of a set of color primaries, the HSV model uses color descriptions that have a more intuitive appeal to a user.
- To give a color specification, a user selects a spectral color and the amounts of white and black that are to be added to obtain different shades, tints, and tones.
- Color parameters in this model are ***hue*** *(H) ,* Saturation (S), and Value (V).

- The three-dimensional representation **of** the HSV model is derived from the RGB **cube.**

R=0
Y=60
G=120
C=180
B=240
M=360



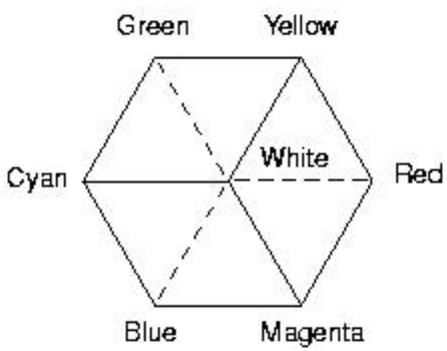
RGB Color Cube
(a)

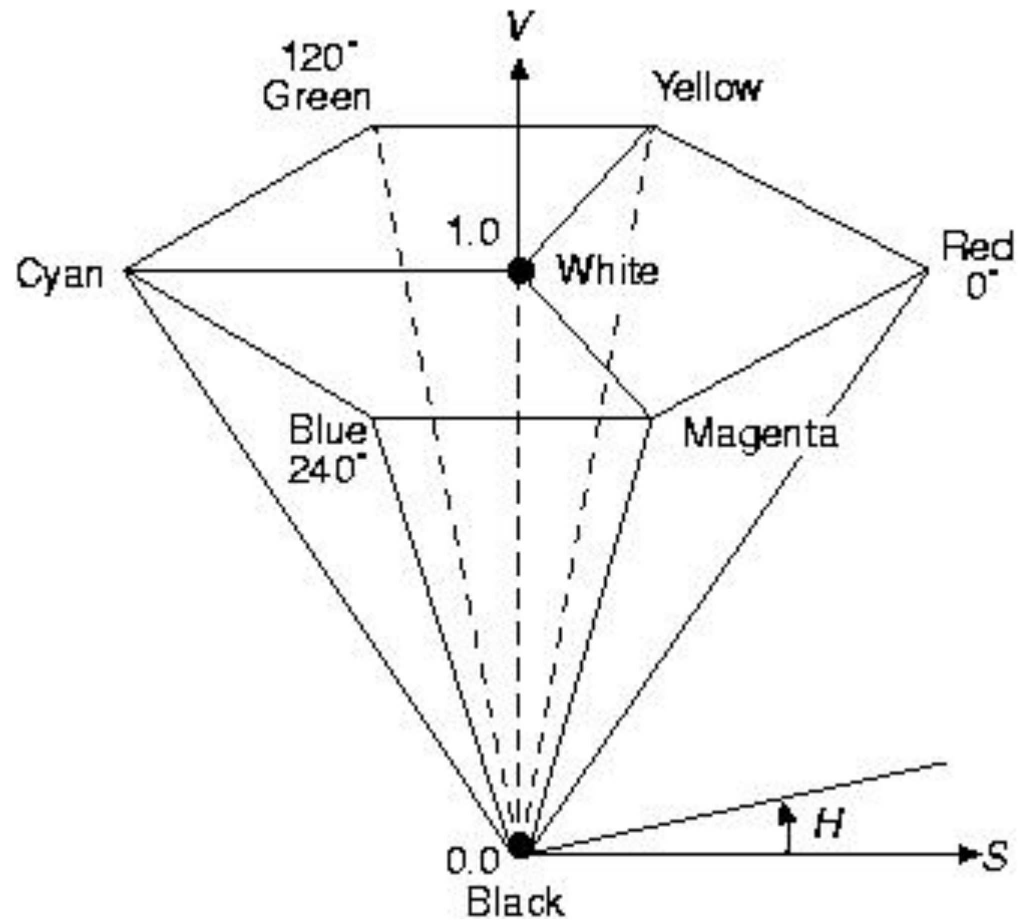


Color Hexagon
(b)

- If we imagine viewing the **cube** along the diagonal from the white vertex to the origin (black), **we see** an outline of the **cube** that has the hexagon shape
- The boundary of the hexagon represents the various hues, and it is used as the top of the HSV hexagon.
- In the hexcone, saturation is measured along a horizontal axis.
- Value is along a vertical axis through the center of the hexcone.

Color Cube Hexcone

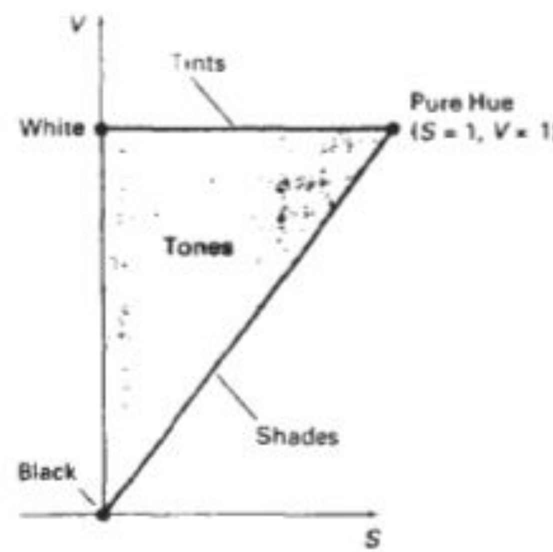




- ***Hue (H) is the angle around the vertical axis***
- ***Saturation (S) is a value from 0 to 1 indicating how far from the vertical axis the color lies***
- ***Value (V) is the height of the hexcone"***

- Hue is represented as an angle about the vertical axis, ranging from 0 at red through **360 .**
- Vertices of the hexagon are separated by 60 intervals.
- Yellow is at 60, green at 120 and **cyan** opposite red at H = **180.**
- Complementary colors are 180 apart.

- Color concepts associated with the terms shades, tints, and tones are represented in a cross-sectional plane of the HSV hexcone
- Adding black to a pure hue decreases V down the side of the hexcone. Thus, various shades are represented with values S - 1 and **0 <= V <=1**



- **Ad**ding white to a pure tone produces different tints across the top plane of the hexcone, where parameter values are V = 1 and **O<=S<=1.**
- Various tones are specified by adding both black and white, producing color points within the triangular cross-sectional area of the hexcone.

- Saturation **S** varies **from 0** to **1.**
- It is represented in this model as the ratio of the purity of a selected hue to its maximum purity at ***S*** = 1.
- A selected hue is said to be one-quarter pure at the value ***S*** = **0.25.** At **S** = **0,** we have the gray scale.

- Value V varies from **0** at the apex of the hexcone to *1* at the top.
- The apex represents black.
- At the top of the hexcone, colors have their maximum intensity.
- When V = 1 and ***S*** = **1,** we have the "pure" hues. White is the point at V = **1** and ***S*** **=o.**

- The human eye can distinguish about **128** different hues and about **130** different tints (saturation levels).
- For each of these, a number of shades (value settings) can be detected, depending on the hue selected.
- About **23** shades are discernible with yellow colors, and about 16 different shades can be **seen** at the blue end of the spectrum.
- This means that we can distinguish about **128 x** 130 x **23** = **82,720** different colors.

- For most graphics applications, **128** hues, 8 saturation levels, and **15** value settings are sufficient. With this range of parameters in the **HSV** color model, **16,384** colors

CONVERSION BETWEEN HSV AND RGB MODELS

- First consider how the HSV hexcone can be derived from the RGB cube. The diagonal of this cube trom black (the origin) to white corresponds to the ***V*** axis of the hexcone.
- Also, each subcube of the RGB cube corresponds to a hexagonal cross-sectional area of the hexcone.
- At any cross section, all sides of the hexagon and all radial lines from the V axis to any vertex have the value V.
- For any set of RGB values, V is equal to the maximum value in this set.
- The HSV point corresponding to the set of RGB values lies on the hexagonal cross section at value V.
- Parameter S is then determined as the relative distance of this point from the V axis.
- Parameter H is determined by calculating the relative position of the point within each sextant of the hexagon.

- Input:RGB
- Output: HSV
- Step1: Normalize the RGB value to be in the range[0,1]
    - r= R/255
    - g=G/255
    - b= B/255
- Step 2: find the difference between max and min values
    - M=max(r,g,b)
    - m=min(r,g,b)
    - Difference Δ=M-m

- Step 3: calculate value V
  - V=max(r,g,b)
- Step 4: calculate saturation S
  - Δ =0,then S=0, otherwise S= Δ/M
- Step 5:Calculate hue H
  - If M=r then H= $60 * \left(\left({}^{g-b}\!\!\diagup_{\Delta} \right) * \bmod 6 \right)$
  - If M=g then H= $60 * \left(2 + \left(b - r \right) / \Delta \right)$
  - If M=b them H= $60 * \left(4 + \left(r - g \right) / \Delta \right)$
  - If H<0 then H=H+360

HSV TO RGB

- **Input: h, s, v in range [O. .1]**
- **Outputs: r , g, b in range [0..1]**
- **if *(S* == 0)** /. **Grayscale** ./
    - **r** = ***g*** = **b** = v
- else
    - – If (h==1) h=0;
    - – H*=6.0
    - – i=floor(h)
    - – f=h-i
    - – aa=v*(1-s)
    - – bb=v*(1-(s*f))
    - – cc=v*(1-(s*(1-f)))

- Case0:r=v,g=cc,b=aa
- Case1:r=bb,g=v,b=aa
- Case2:r=aa,g=v,b=cc
- Case3:r=aa,g=bb,b=v
- Case4:r=cc,g=aa,b=v
- Case5:r=v,g=aa,b=bb

- RGB value(29,98,128) convert into CMY

$R=29/255 \ =0.113$

$G=98/255 \ =0.384$

$B=128/255= 0.501$

$C=B+G =W-R$

$M=R+B=W-G$

$Y=G+R=W-B$

$C=1-0.113=0.887$

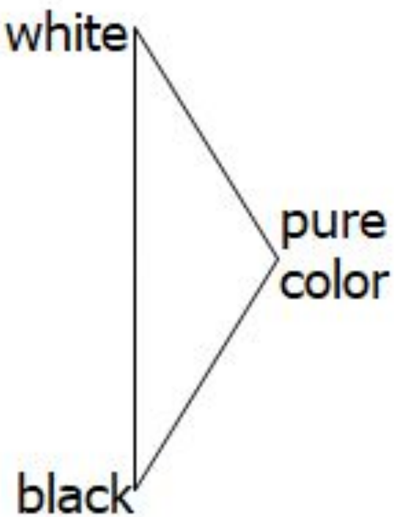
$M=1-0.384=0.616$

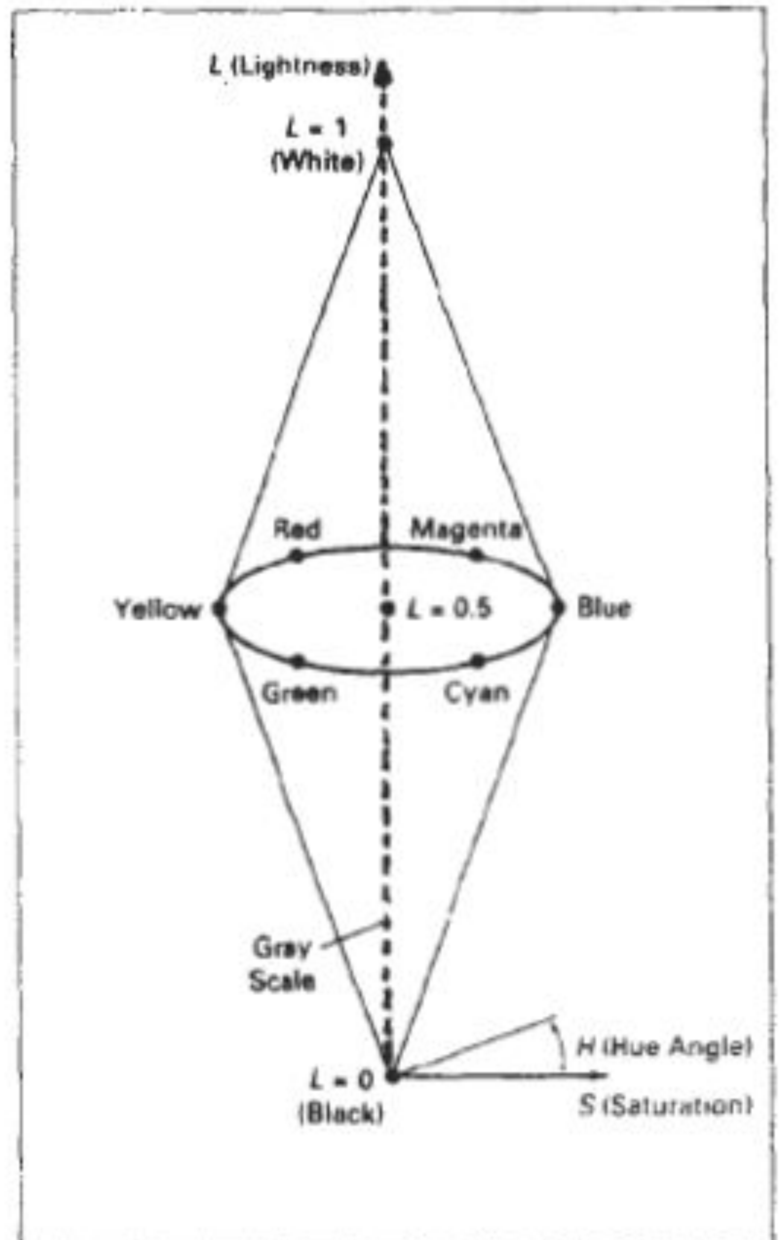
$Y=1-0.501=0.499$

HLS COLOR MODEL

- This model has the double-cone representation
- The three color parameters in this model are called ***hue*** *(H),* ***lightness (L),*** and ***saturation*** *(S)*
- The vertical axis in this model is called lightness, L.
- At L = 0, we have black, and white is at L = 1. Gray scale is along the L axis, and the "pure hues" lie on the L = 0.5 plane.
- Saturation parameter S again specifies relative purity of a color.
- This parameter varies from 0 to 1, and pure hues are those for which S = 1 and **L** = 0.5.
- As S decreases, the hues are said to be less pure. At S = 0, we have the gray scale.

- Double cones





- Hue has the same meaning as in the HSV model.
- It specifies an angle about the vertical axis that locates a chosen hue.
- In this model, H = 0 corresponds to blue.
- Magenta is at ***60*** red is at 120 and cyan is located
- at H = 180 . Again, complementary colors are 180 apart on the double cone.
- A hue is selected with hue angle H, and the desired shade, tint, or tone is obtained by adjusting L and S.
- Colors are made lighter by increasing L and made darker by decreasing L.
- When S is decreased, the colors move toward gray.

- Shades lie at bottom half of the boundary.
- Tint lie at upper half of the boundary.
- Tones lie inside the entire double hexacone.